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Event Management System

Case Study 119 · C++ Programming · Semester II

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Note: Content slides (excluding Title & Thank You) are limited to 4.

01 Problem Statement

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THE PROBLEM

Managing event registrations manually is time-consuming and error-prone.

- Overbooking beyond venue capacity
- Duplicate participant entries
- No way to cancel a registration
- No real-time count of participants
- Data lost when the program closes

WHAT WE NEED

- Register participants
- Cancel registrations
- Check event capacity
- Prevent duplicate entries
- Save data permanently
- Display event summary

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02 Our Solution

Event Management System built in C++ using File Handling

1 Register Participant

Enter name & phone number. System checks if event is full and prevents duplicate phone numbers before saving to file.

2 Cancel Registration

Enter name & phone to remove. Uses a temporary file method to safely delete the exact record without losing other data.

3 View Event Details

Displays event name, date, total capacity, current participant count, and remaining available slots.

4 View Participants

Shows complete list of all registered participants with serial numbers, names, phone numbers and total count.

03

Working / Method Flow

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1 Program Starts
Global vars set

2 Menu Shown
User sees options

3 Choice Made
Input read

4 Function Runs
Logic executes

5 File Updated
Data saved

KEY CONCEPTS USED IN THIS PROGRAM

File Handling
ifstream & **ofstream** to read/write participant data to participants.txt

getline()
Splits each file line into name and phone number at the last space

rfind(' ')
Reads full names with spaces (e.g. 'Rahul Sharma') correctly

Temp File Delete
Rewrites entire file skipping the cancelled participant's record

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04 Conclusion

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WHAT WE ACHIEVED

- All case study requirements fulfilled
- Participants can register, cancel & be viewed
- Data saved in file — persists on restart
- Capacity validation prevents overbooking
- Duplicate phone check prevents repeat entries
- Clean menu-driven interface works correctly

WHAT I LEARNED

- File Handling (ifstream/ofstream)
- Organizing code using functions
- Input validation techniques
- String parsing with rfind/substr
- Debugging and fixing real bugs
- Writing clean, readable C++

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Thank You

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